

# US Patent & Trademark Office

## Patent Public Search | Text View

United States Patent

12408371

Kind Code

B2

Date of Patent

September 02, 2025

Inventor(s)

Hsiung; Chih-Wen et al.

### NMOS half-bridge power device and manufacturing method thereof

#### Abstract

An NMOS half-bridge power device includes: a semiconductor layer, a plurality of insulation regions, a first N-type high voltage well and a second N-type high voltage well, which are formed by one same ion implantation process, a first P-type high voltage well and a second P-type high voltage well, which are formed by one same ion implantation process, a first drift oxide region and a second drift oxide region, which are formed by one same etch process including etching a drift oxide layer; a first gate and a second gate, which are formed by one same etch process including etching a poly silicon layer, a first P-type body region and a second P-type body region, which are formed by one same ion implantation process, a first N-type source and a first N-type drain, and a second N-type source and a second N-type drain.

**Inventors:** Hsiung; Chih-Wen (Hsinchu, TW), Weng; Wu-Te (Hsinchu, TW), Yang; Ta-Yung (Taoyuan, TW)

**Applicant:** Richtek Technology Corporation (HsinChu, TW)

**Family ID:** 86608123

**Assignee:** RICHTEK TECHNOLOGY CORPORATION (Zhubei, TW)

**Appl. No.:** 17/983434

**Filed:** November 09, 2022

#### Prior Publication Data

**Document Identifier**

**Publication Date**

US 20230178648 A1

Jun. 08, 2023

#### Foreign Application Priority Data

TW

111116926

May. 05, 2022

Related U.S. Application Data

us-provisional-application US 63264934 20211203

Publication Classification

Int. Cl.: H03F3/217 (20060101); H02M3/335 (20060101); H02M7/219 (20060101); H10D30/01 (20250101); H10D30/60 (20250101); H10D30/65 (20250101); H10D62/17 (20250101); H10D64/27 (20250101)

U.S. Cl.:

CPC H10D30/65 (20250101); H03F3/2173 (20130101); H10D30/0221 (20250101); H10D30/603 (20250101); H10D62/393 (20250101); H10D64/516 (20250101); H02M3/33571 (20210501); H02M7/219 (20130101)

Field of Classification Search

CPC: H10D (30/65); H10D (30/0221); H10D (30/603); H10D (62/393); H10D (64/516); H10D (30/0281); H10D (62/371); H10D (84/0151); H10D (84/0156); H10D (84/038); H10D (84/83); H10D (84/859); H10D (84/0188); H10D (62/116); H10D (84/017); H10D (84/0191); H10D (84/85); H01L (21/761); H02P (7/04); H02M (3/33571); H02M (1/0085); H02M (1/4233); H02M (7/5387); H02M (7/219); H03F (3/2173)

References Cited

U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name | U.S. Cl.    | CPC          |
|--------------|-------------|---------------|-------------|--------------|
| 4910563      | 12/1989     | Tuska         | 257/329     | H10D 84/85   |
| 6744649      | 12/2003     | Yang          | 363/41      | H02M 3/3376  |
| 7459750      | 12/2007     | Ludikhuize    | 257/334     | H10D 84/811  |
| 8063444      | 12/2010     | Chang         | 257/E27.06  | H10D 30/603  |
| 8193559      | 12/2011     | Haeberlen     | 257/773     | H10D 30/664  |
| 8288820      | 12/2011     | Kocon         | 327/564     | H10D 30/603  |
| 8293612      | 12/2011     | Lee           | 438/296     | H10D 84/0188 |
| 9093914      | 12/2014     | Tseng         | N/A         | H02M 3/07    |
| 9614074      | 12/2016     | Lin           | N/A         | H10D 30/603  |
| 10249637     | 12/2018     | Torii         | N/A         | H10D 64/685  |
| 10461624     | 12/2018     | Tsu           | N/A         | H02M 1/08    |
| 11935950     | 12/2023     | Chen          | N/A         | H10D 30/0221 |
| 11967644     | 12/2023     | Wang          | N/A         | H10D 62/107  |
| 2004/0125619 | 12/2003     | Yang          | 363/21.01   | H02M 3/3376  |
| 2005/0024897 | 12/2004     | Yang          | 363/21.06   | H02M 1/38    |
| 2005/0199962 | 12/2004     | Suzuki        | 257/368     | H10D 30/0221 |
| 2005/0280998 | 12/2004     | Lin           | 361/704     | H05K 7/209   |
| 2006/0006462 | 12/2005     | Chang         | 257/E21.549 | H10D 62/116  |
| 2006/0054967 | 12/2005     | Ludikhuize    | 257/E27.06  | H10D 84/856  |
| 2008/0309256 | 12/2007     | Wang          | 315/307     | H05B 41/2828 |

|              |         |             |             |               |
|--------------|---------|-------------|-------------|---------------|
| 2010/0001345 | 12/2009 | Furuhata    | 257/E29.261 | H10D 84/017   |
| 2010/0187606 | 12/2009 | Kobayashi   | 257/E29.256 | H10D 84/017   |
| 2010/0315159 | 12/2009 | Kocon       | 257/E21.409 | H10D 30/0281  |
| 2011/0241170 | 12/2010 | Haeberlen   | 257/532     | H10D 84/038   |
| 2013/0015523 | 12/2012 | You         | 438/286     | H10D 30/65    |
| 2013/0200453 | 12/2012 | Park        | 257/338     | H10D 84/401   |
| 2013/0341717 | 12/2012 | Chen et al. | N/A         | N/A           |
| 2014/0001549 | 12/2013 | Bode        | 438/478     | H10D 30/0285  |
| 2014/0118029 | 12/2013 | Tseng       | 327/79      | H03K 5/1536   |
| 2014/0191732 | 12/2013 | Tseng       | 320/166     | H02M 1/36     |
| 2014/0320174 | 12/2013 | Lu          | 327/109     | H10D 30/65    |
| 2016/0099349 | 12/2015 | Parris      | 438/286     | H10D 62/107   |
| 2017/0179279 | 12/2016 | Lin         | N/A         | H10D 30/0221  |
| 2017/0200785 | 12/2016 | Janssens    | N/A         | H01L 21/76224 |
| 2018/0138797 | 12/2017 | Tsu         | N/A         | H02M 3/33523  |
| 2018/0174887 | 12/2017 | Chuang      | N/A         | H10D 62/393   |
| 2018/0240876 | 12/2017 | Tu et al.   | N/A         | N/A           |
| 2018/0261617 | 12/2017 | Torii       | N/A         | H10D 64/685   |
| 2018/0277531 | 12/2017 | Takeuchi    | N/A         | H10D 62/127   |
| 2019/0081045 | 12/2018 | Lin         | N/A         | H10D 30/65    |
| 2021/0175165 | 12/2020 | Shibuya     | N/A         | H01L 25/072   |
| 2023/0081508 | 12/2022 | Tu          | 257/288     | H10D 30/603   |
| 2023/0178438 | 12/2022 | Weng        | 438/294     | H10D 84/0135  |
| 2023/0197730 | 12/2022 | Weng        | 257/338     | H10D 30/65    |
| 2024/0106342 | 12/2023 | Hu          | N/A         | H01L 24/73    |
| 2024/0113115 | 12/2023 | Naik        | N/A         | H10D 30/0289  |
| 2024/0194679 | 12/2023 | Bienvenu    | N/A         | H10D 84/8312  |
| 2024/0339522 | 12/2023 | Xu          | N/A         | H10D 84/038   |

*Primary Examiner:* Malek; Maliheh

*Attorney, Agent or Firm:* Tung & Associates

## Background/Summary

CROSS REFERENCE (1) The present invention claims priority to US 63/264,934 filed on Dec. 3, 2021 and claims priority to TW 11116926 filed on May 5, 2022.

### BACKGROUND OF THE INVENTION

#### Field of Invention

(1) The present invention relates to an NMOS half-bridge power device and a manufacturing method thereof; particularly, it relates to an NMOS half-bridge power device integrating an NMOS upper gate device and an NMOS lower gate device and a manufacturing method thereof.

#### Description of Related Art

(2) Conventionally, a buck power stage circuit comprises a half-bridge switching circuit constituted by an upper gate power device and a lower gate power device. The upper gate power device and the lower gate power device are separately formed by different manufacturing processes, which has the drawbacks of limited application range and high manufacturing cost.

(3) In view of the above, to overcome the drawbacks in the prior art, the present invention proposes

an NMOS half-bridge power device integrating an NMOS upper gate device and an NMOS lower gate device therein, which can be manufactured by an integration process.

## SUMMARY OF THE INVENTION

(4) From one perspective, the present invention provides an NMOS half-bridge power device, comprising: a semiconductor layer, which is formed on a substrate; a plurality of insulation regions, which are formed on the semiconductor layer, for defining an NMOS upper gate device region and an NMOS lower gate device region, wherein an NMOS upper gate device is formed in the NMOS upper gate device region, whereas, an NMOS lower gate device is formed in the NMOS lower gate device region; a first N-type buried layer, which is formed in the NMOS upper gate device region; a first N-type high voltage isolation region and a second N-type high voltage isolation region, which are formed in the semiconductor layer of the NMOS upper gate device region by one same ion implantation process; a first N-type high voltage well and a second N-type high voltage well, which are formed in the semiconductor layer of the NMOS upper gate device region and the semiconductor layer of the NMOS lower gate device region, respectively, by one same ion implantation process; a first P-type high voltage well and a second P-type high voltage well, which are formed in the semiconductor layer of the NMOS upper gate device region and the semiconductor layer of the NMOS lower gate device region, respectively, by one same ion implantation process; wherein the first N-type high voltage well and the first P-type high voltage well are in contact with each other in a channel direction, and wherein the second N-type high voltage well and the second P-type high voltage well are in contact with each other in the channel direction; a first drift oxide region and a second drift oxide region, which are formed, by one same etching process including etching a drift oxide layer, in the NMOS upper gate device region and in the NMOS lower gate device region, respectively; a first gate and a second gate, which are formed, by one same etching process including etching a polysilicon layer, in the NMOS upper gate device region and in the NMOS lower gate device region, respectively; a first P-type body region and a second P-type body region, which are formed in the semiconductor layer of the NMOS upper gate device region and the semiconductor layer of the NMOS lower gate device region, respectively, by one same ion implantation process; wherein a portion of the first P-type body region is located vertically below the first gate, and wherein the first P-type body region and the first N-type high voltage well are in contact with each other in the channel direction, wherein a portion of the second P-type body region is located vertically below the second gate, and wherein the second P-type body region and the second N-type high voltage well are in contact with each other in the channel direction; a first N-type source and a first N-type drain, which are formed, by one same ion implantation process, in the semiconductor layer of the NMOS upper gate device region, wherein the first N-type source and the first N-type drain are located below and outside two sides of the first gate, respectively, wherein a side of the first gate which is closer to the first N-type source is a source side and another side of the first gate which is closer to the first N-type drain is a drain side, and wherein the first N-type source is located in the first P-type body region, and the first N-type drain is located in the first N-type high voltage well; and a second N-type source and a second N-type drain, which are formed in the semiconductor layer of the NMOS lower gate device region by the one same ion implantation process that forms the first N-type source and the first N-type drain, wherein the second N-type source and the second N-type drain are located below and outside two sides of the second gate, respectively, wherein a side of the second gate which is closer to the second N-type source is a source side and another side of the second gate which is closer to the second N-type drain is a drain side, and wherein the second N-type source is located in the second P-type body region, and the second N-type drain is located in the second N-type high voltage well; wherein the first N-type buried layer is formed in the semiconductor layer and in the substrate and is vertically below the first N-type high voltage well and the first P-type high voltage well; wherein in the channel direction, the first N-type high voltage isolation region is in contact with a side of the first N-type high voltage well, wherein this side of the first N-type high voltage well is opposite

to another side of the first N-type high voltage well which is in contact with the first P-type high voltage well; wherein in the channel direction, the second N-type high voltage isolation region is in contact with a side of the first P-type high voltage well, wherein this side of the first P-type high voltage well is opposite to another side of the first P-type high voltage well which is in contact with the first N-type high voltage well.

(5) From another perspective, the present invention provides a manufacturing method of an NMOS half-bridge power device, wherein the NMOS half-bridge power device includes: an NMOS upper gate device and an NMOS lower gate device; the manufacturing method of the NMOS half-bridge power device comprising: forming a semiconductor layer on a substrate; forming a plurality of insulation regions on the semiconductor layer, for defining an NMOS upper gate device region and an NMOS lower gate device region, wherein an NMOS upper gate device is formed in the NMOS upper gate device region, whereas, an NMOS lower gate device is formed in the NMOS lower gate device region; forming a first N-type buried layer in the NMOS upper gate device region; forming a first N-type high voltage isolation region and a second N-type high voltage isolation region in the semiconductor layer of the NMOS upper gate device region by one same ion implantation process; forming a first N-type high voltage well and a second N-type high voltage well in the semiconductor layer of the NMOS upper gate device region and the semiconductor layer of the NMOS lower gate device region, respectively, by one same ion implantation process; forming a first P-type high voltage well and a second P-type high voltage well in the semiconductor layer of the NMOS upper gate device region and the semiconductor layer of the NMOS lower gate device region, respectively, by one same ion implantation process; wherein the first N-type high voltage well and the first P-type high voltage well are in contact with each other in a channel direction, and wherein the second N-type high voltage well and the second P-type high voltage well are in contact with each other in the channel direction; forming a drift oxide layer on the semiconductor layer, wherein the drift oxide layer overlays the NMOS upper gate device region and the NMOS lower gate device region; etching the drift oxide layer to form a first drift oxide region in the NMOS upper gate device region and to form a second drift oxide region in the NMOS lower gate device region by one same etching process; subsequent to the formation of the first drift oxide region and the second drift oxide region, forming a gate dielectric layer on the semiconductor layer, wherein the gate dielectric layer overlays the NMOS upper gate device region and the NMOS lower gate device region; forming a polysilicon layer on the gate dielectric layer, wherein the polysilicon layer overlays the NMOS upper gate device region and the NMOS lower gate device region; etching the polysilicon layer to form a first gate in the NMOS upper gate device region and to form a second gate in the NMOS lower gate device region by one same etching process; forming a first P-type body region and a second P-type body region in the semiconductor layer of the NMOS upper gate device region and the semiconductor layer of the NMOS lower gate device region, respectively, by one same ion implantation process; wherein a portion of the first P-type body region is located vertically below the first gate, and wherein the first P-type body region and the first N-type high voltage well are in contact with each other in the channel direction, wherein a portion of the second P-type body region is located vertically below the second gate, and wherein the second P-type body region and the second N-type high voltage well are in contact with each other in the channel direction; forming a first N-type source and a first N-type drain in the semiconductor layer of the NMOS upper gate device region by one same ion implantation process, wherein the first N-type source and the first N-type drain are located below and outside two sides of the first gate, respectively, wherein a side of the first gate which is closer to the first N-type source is a source side and another side of the first gate which is closer to the first N-type drain is a drain side, and wherein the first N-type source is located in the first P-type body region, and the first N-type drain is located in the first N-type high voltage well; and forming a second N-type source and a second N-type drain in the semiconductor layer of the NMOS lower gate device region by the one same ion implantation process that forms the first N-type source and the first N-type drain, wherein the

second N-type source and the second N-type drain are located below and outside two sides of the second gate, respectively, wherein a side of the second gate which is closer to the second N-type source is a source side and another side of the second gate which is closer to the second N-type drain is a drain side, and wherein the second N-type source is located in the second P-type body region, and the second N-type drain is located in the second N-type high voltage well; wherein the first N-type buried layer is formed in the semiconductor layer and in the substrate and is vertically below the first N-type high voltage well and the first P-type high voltage well; wherein in the channel direction, the first N-type high voltage isolation region is in contact with a side of the first N-type high voltage well, wherein this side of the first N-type high voltage well is opposite to another side of the first N-type high voltage well which is in contact with the first P-type high voltage well; wherein in the channel direction, the second N-type high voltage isolation region is in contact with a side of the first P-type high voltage well, wherein this side of the first P-type high voltage well is opposite to another side of the first P-type high voltage well which is in contact with the first N-type high voltage well.

(6) In one embodiment, the NMOS half-bridge power device further comprises: a first P-type conductive region, which is formed in the first P-type body region, wherein the first P-type conductive region serves as an electrical contact of the first P-type body region; and a second P-type conductive region, which is formed in the second P-type body region by the one same ion implantation process that forms the first P-type conductive region, wherein the second P-type conductive region serves as an electrical contact of the second P-type body region.

(7) In one embodiment, the NMOS half-bridge power device further comprises: a second N-type buried layer, which is formed in the NMOS lower gate device region; and a third N-type high voltage isolation region and a fourth N-type high voltage isolation region, which are formed in the semiconductor layer of the NMOS lower gate device region by the one same ion implantation process that forms the first N-type high voltage isolation region and the second N-type high voltage isolation region; wherein the second N-type buried layer is formed in the semiconductor layer and in the substrate and is vertically below the second N-type high voltage well and the second P-type high voltage well; wherein in the channel direction, the third N-type high voltage isolation region is in contact with a side of the second N-type high voltage well, wherein the side of the second N-type high voltage well is opposite to another side of the second N-type high voltage well which is in contact with the second P-type high voltage well; wherein in the channel direction, the fourth N-type high voltage isolation region is in contact with a side of the second P-type high voltage well, wherein the side of the second P-type high voltage well is opposite to another side of the second P-type high voltage well which is in contact with the second N-type high voltage well.

(8) In one embodiment, the first N-type source is electrically connected to the second N-type source, and the first P-type conductive region is electrically connected to the second N-type source.

(9) In one embodiment, the first gate of the NMOS upper gate device has a gate length of  $0.4 \sim 1 \mu\text{m}$ , wherein a length of a portion where the first gate overlays the first drift oxide region is  $0.1 \sim 0.4 \mu\text{m}$ .

(10) In one embodiment, the second gate of the NMOS lower gate device has a gate length of  $0.4 \sim 1 \mu\text{m}$ , wherein a length of a portion where the second gate overlays the second drift oxide region is  $0.1 \sim 0.4 \mu\text{m}$ .

(11) In one embodiment, the semiconductor layer is a P-type semiconductor epitaxial layer having a volume resistivity of  $45 \text{ Ohm-cm}$ .

(12) In one embodiment, each of the first drift oxide region and the second drift oxide region has a thickness ranging between  $400 \text{ \AA}$  and  $450 \text{ \AA}$ .

(13) In one embodiment, each of the dielectric layer of the first gate and the dielectric layer of the second gate has a thickness ranging between  $80 \text{ \AA}$  and  $100 \text{ \AA}$ .

(14) In one embodiment, a gate driving voltage of the NMOS upper gate device region is  $3.3\text{V}$ , and the first N-type drain is electrically connected to a voltage level ranging between  $6\text{V}$  and

(15) In one embodiment, the NMOS half-bridge power device has a minimum feature size of  $0.18$

μm.

(16) Advantages of the present invention include: that, the present invention can form different units of the NMOS upper gate device and the NMOS lower gate device of the NMOS half-bridge switching circuit at the same time by one same manufacturing process to integrate the NMOS half-bridge switching circuit into one NMOS half-bridge power device; and that, the present invention forms an isolation region in the semiconductor layer to electrically isolate the NMOS upper gate device and the NMOS lower gate device.

(17) The objectives, technical details, features, and effects of the present invention will be better understood with regard to the detailed description of the embodiments below, with reference to the attached drawings.

---

## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 shows a cross-section view of an NMOS half-bridge power device according to an embodiment of the present invention.

(2) FIG. 2 shows a cross-section view of an NMOS half-bridge power device according to an embodiment of the present invention.

(3) FIG. 3A to FIG. 3N show cross-section views of a manufacturing method of an NMOS half-bridge power device according to an embodiment of the present invention.

### DESCRIPTION OF THE PREFERRED EMBODIMENTS

(4) The drawings as referred to throughout the description of the present invention are for illustration only, to show the interrelations among the process steps and the layers, while the shapes, thicknesses, and widths are not drawn in actual scale.

(5) Please refer to FIG. 1, which shows a cross-section view of an NMOS half-bridge power device according to an embodiment of the present invention. As shown in FIG. 1, the NMOS half-bridge power device **10** comprises: a semiconductor layer **11'**, insulation regions **12**, a first N-type buried layer **13a**, a first N-type high voltage isolation region **14c** and a second N-type high voltage isolation region **14d** which are formed by one same ion implantation process, a first N-type high voltage well **14a** and a second N-type high voltage well **14b** which are formed by one same ion implantation process, a first P-type high voltage well **15a** and a second P-type high voltage well **15b** which are formed by one same ion implantation process, a first drift oxide region **16a** and a second drift oxide region **16b** which are formed by one same etching process including etching a drift oxide layer, a first gate **17a** and a second gate **17b** which are formed by one same etching process including etching a polysilicon layer, a first P-type body region **19a** and a second P-type body region **19b** which are formed by one same ion implantation process, a first N-type source **18a** and a first N-type drain **18b**, and a second N-type source **18c** and a second N-type drain **18d**.

(6) A semiconductor layer **11'** is formed on the substrate **11**. The semiconductor layer **11'** has a top surface **11a** and a bottom surface **11b** opposite to the top surface **11a** in a vertical direction (as indicated by the direction of the solid arrow in FIG. 1, and all occurrences of the term “vertical direction” in this specification refer to the same direction). The semiconductor layer **11'**, for example, is formed on the substrate **11** by an epitaxial process, or is a part of the substrate **11**. The semiconductor layer **11'** can be formed by various methods known to a person having ordinary skill in the art, so the details thereof are not redundantly explained here.

(7) Please still refer to FIG. 1. Plural insulation regions **12** are formed on the semiconductor layer **11'**, for defining an NMOS upper gate device region HS-NMOS and an NMOS lower gate device region LS-NMOS, wherein an NMOS upper gate device **10a** is formed in the NMOS upper gate device region HS-NMOS, whereas, an NMOS lower gate device **10b** is formed in the NMOS lower gate device region LS-NMOS. The insulation regions **12** can be, for example but not limited to, a

shallow trench isolation (STI) structure shown in FIG. 1.

(8) In this embodiment, the NMOS upper gate device **10a** includes: a first N-type buried layer **13a**, a first N-type high voltage isolation region **14c**, a second N-type high voltage isolation region **14d**, a first N-type high voltage well **14a**, a first P-type high voltage well **15a**, a first drift oxide region **16a**, a first gate **17a**, a first N-type source **18a**, a first N-type drain **18b**, a first P-type body region **19a** and a first P-type conductive region **19c**. The NMOS lower gate device **10b** includes: a second N-type high voltage well **14b**, a second P-type high voltage well **15b**, a second drift oxide region **16b**, a second gate **17b**, a second N-type source **18c**, a second N-type drain **18d**, a second P-type body region **19b**, a second P-type conductive region **19d**.

(9) Please still refer to FIG. 1. The first N-type buried layer **13a** is formed in the NMOS upper gate device region HS-NMOS. The first N-type buried layer **13a** is formed in the semiconductor layer **11'** and in the substrate **11** and is vertically below the first N-type high voltage well **14a** and the first P-type high voltage well **15a**. The first N-type high voltage isolation region **14c** and the second N-type high voltage isolation region **14d** are formed in the semiconductor layer **11'** of the NMOS upper gate device region HS-NMOS by one same ion implantation process. In a channel direction (as indicated by the direction of the dashed arrow shown in FIG. 1, and all occurrences of the term "channel direction" in this specification refer to the same direction), the first N-type high voltage isolation region **14c** is in contact with a side of the first N-type high voltage well **14a**, wherein this side of the first N-type high voltage well **14a** is opposite to another side of the first N-type high voltage well **14a** which is in contact with the first P-type high voltage well **15a**. In the channel direction, the second N-type high voltage isolation region **14d** is in contact with a side of the first P-type high voltage well **15a**, wherein this side of the first P-type high voltage well **15a** is opposite to another side of the first P-type high voltage well **15a** which is in contact with the first N-type high voltage well **14a**. The first N-type buried layer **13a**, the first N-type high voltage isolation region **14c** and the second N-type high voltage isolation region **14d** together constitute a device isolation region, wherein such device isolation region entirely encompasses the lower boundary of the NMOS upper gate device **10a** below the top surface **11a** of the semiconductor layer **11'**, so that below the top surface **11a** of the semiconductor layer **11'**, the NMOS upper gate device **10a** is electrically isolated from other devices which are formed in the semiconductor layer **11'**, such as the NMOS lower gate device **10b**.

(10) Please still refer to FIG. 1. The first N-type high voltage well **14a** and the second N-type high voltage well **14b** are formed in the semiconductor layer **11'** of the NMOS upper gate device region HS-NMOS and the semiconductor layer **11'** of the NMOS lower gate device region LS-NMOS, respectively, by one same ion implantation process. Both the first N-type high voltage well **14a** and the second N-type high voltage well **14b** are below and in contact with the top surface **11a**. A part of the first N-type high voltage well **14a** is located vertically below and in contact with the first gate **17a**, which serve as a drift current channel in an ON operation of the NMOS upper gate device **10a**. Besides, a part of the second N-type high voltage well **14b** is located vertically below and in contact with the second gate **17b**, which serve as a drift current channel in an ON operation of the NMOS lower gate device **10b**.

(11) Please still refer to FIG. 1. The first P-type high voltage well **15a** and the second P-type high voltage well **15b** are formed in the semiconductor layer **11'** of the NMOS upper gate device region HS-NMOS and the semiconductor layer **11'** of the NMOS lower gate device region LS-NMOS, respectively, by one same ion implantation process. The first N-type high voltage well **14a** and the first P-type high voltage well **15a** are in contact with each other in the channel direction. The second N-type high voltage well **14b** and the second P-type high voltage well **15b** are in contact with each other in the channel direction. Both the first P-type high voltage well **15a** and the second P-type high voltage well **15b** are below and in contact with the top surface **11a**.

(12) The first drift oxide region **16a** and the second drift oxide region **16b** are formed, by one same etching process including etching a drift oxide layer, in the NMOS upper gate device region HS-



NMOS and in the NMOS lower gate device region LS-NMOS, respectively. The first drift oxide region **16a** and the second drift oxide region **16b** are formed on the semiconductor layer **11'**, and are located a drift region of the NMOS upper gate device **10a** and a drift region of the NMOS lower gate device **10b**, respectively.

(13) The first gate **17a** and a second gate **17b** are formed, by one same etching process including etching a polysilicon layer, in the NMOS upper gate device region HS-NMOS and in the NMOS lower gate device region LS-NMOS, respectively.

(14) The first gate **17a** and the second gate **17b** are formed on the top surface **11a** of the semiconductor layer **11'**. Each of the first gate **17a** and the second gate **17b** includes: a conductive layer, a spacer layer and a dielectric layer, wherein the dielectric layer is located on and in contact with the top surface **11a**, which is known to a person having ordinary skill in the art, so the details thereof are not redundantly explained here.

(15) Please still refer to FIG. 1. The first P-type body region **19a** and the second P-type body region **19b** are formed in the semiconductor layer **11'** of the NMOS upper gate device region HS-NMOS and the semiconductor layer **11'** of the NMOS lower gate device region LS-NMOS, respectively, by one same ion implantation process. The first P-type body region **19a** and the first N-type high voltage well **14a** are in contact with each other in the channel direction. The second P-type body region **19b** and the second N-type high voltage well **14b** are in contact with each other in the channel direction.

(16) A portion of the first P-type body region **19a** is located vertically below and in contact with the first gate **17a**, which serve as an inversion current channel in an ON operation of the NMOS upper gate device **10a**. And, a portion of the second P-type body region **19b** is located vertically below and in contact with the second gate **17b**, which serve as an inversion current channel in an ON operation of the NMOS lower gate device **10b**.

(17) The first N-type source **18a** and the first N-type drain **18b** are formed, by one same ion implantation process, in the semiconductor layer **11'** of the NMOS upper gate device region HS-NMOS, wherein the first N-type source **18a** and the first N-type drain **18b** are located below and outside two sides of the first gate **17a** in the channel direction, respectively, wherein the side of the first gate **17a** which is closer to the first N-type source **18a** is a source side and the side of the first gate **17a** which is closer to the first N-type drain **18b** is a drain side, and wherein the first N-type source **18a** is located in the first P-type body region **19a**, and the first N-type drain **18b** is located in the first N-type high voltage well **14a**.

(18) In the vertical direction, the first N-type source **18a** and the first N-type drain **18b** are formed on and in contact with the top surface **11a**. Besides, in the channel direction, a drift region of the NMOS upper gate device **10a** lies between the first N-type drain **18b** and the first P-type body region **19a**, so as to separate the first N-type drain **18b** from the first P-type body region **19a**. And, a position in the first N-type high voltage well **14a** which is near the top surface **11a** serves as a drift current channel in an ON operation of the NMOS upper gate device **10a**.

(19) The second N-type source **18c** and a second N-type drain **18d** are formed in the semiconductor layer **11'** of the NMOS lower gate device region LS-NMOS by one same ion implantation process, wherein the second N-type source **18c** and the second N-type drain **18d** are located below and outside two sides of the second gate **17b** in the channel direction, respectively, wherein the side of the second gate **17b** which is closer to the second N-type source **18c** is a source side and the side of the second gate **17b** which is closer to the second N-type drain **18d** is a drain side, and wherein the second N-type source **18c** is located in the second P-type body region **19b**, and the second N-type drain **18d** is located in the second N-type high voltage well **14b**.

(20) In the vertical direction, the second N-type source **18c** and the second N-type drain **18d** are formed on and in contact with the top surface **11a**. Besides, in the channel direction, a drift region of the NMOS lower gate device **10b** lies between the second N-type drain **18d** and the second P-type body region **19b**, so as to separate the second N-type drain **18d** from the second P-type body

region **19b**. And, a position of the second N-type high voltage well **14b** which is near the top surface **11a** serves as a drift current channel in an ON operation of the NMOS lower gate device **10b**.

(21) As shown in FIG. 1, the first P-type conductive region **19c** is formed in the first P-type body region **19a**, wherein the first P-type conductive region **19c** serves as an electrical contact of the first P-type body region **19a**. The second P-type conductive region **19d** is formed in the second P-type body region **19c** by the one same ion implantation process that forms the first P-type conductive region **19c**, wherein the second P-type conductive region **19c** serves as an electrical contact of the second P-type body region **19b**.

(22) The first P-type conductive region **19c** and the second P-type conductive region **19d** are formed on and in contact with the top surface **11a**. Besides, in the channel direction, the first P-type conductive region **19c** is in contact with and electrically connected to the first N-type source **18a**, whereas, the second P-type conductive region **19d** is in contact with and electrically connected to the second N-type source **18c**. As shown in FIG. 1, the first P-type conductive region **19c**, the first N-type source **18a** and the second N-type drain **18d** are electrically connected to a switch node voltage SW. In addition, the second P-type conductive region **19d** and the second N-type source **18c** are electrically connected to ground. The first N-type drain **18b** is electrically connected to an input voltage  $V_{in}$ . In a case wherein the NMOS half-bridge power device is applied to a buck power stage circuit, the switch node is electrically connected to one end of an inductor, whereas, another end of the inductor is electrically connected to an output voltage. A buck power stage circuit is known to a person having ordinary skill in the art, so the details thereof are not redundantly explained here.

(23) In one embodiment, the first gate **17a** of the NMOS upper gate device **10a** has a gate length of  $0.4 \sim 1 \mu\text{m}$ . And, a length of a portion where the first gate **17a** overlays the first drift oxide region **16a** is  $0.1 \sim 0.4 \mu\text{m}$ .

(24) In one embodiment, the second gate **17b** of the NMOS lower gate device **10b** has a gate length of  $0.4 \sim 1 \mu\text{m}$ . And, a length of a portion where the second gate **17b** overlays the second drift oxide region **16b** is  $0.1 \sim 0.4 \mu\text{m}$ .

(25) In one embodiment, the semiconductor layer **11'** is a P-type semiconductor epitaxial layer having a volume resistivity of 45 Ohm-cm.

(26) In one embodiment, each of the first drift oxide region **16a** and the second drift oxide region **16b** is a chemical vapor deposition (CVD) oxide region.

(27) In one embodiment, each of the first drift oxide region **16a** and the second drift oxide region **16b** has a thickness ranging between 400 Å to 450 Å.

(28) In one embodiment, each of the dielectric layer of the first gate **17a** and the dielectric layer of the second gate **17b** has a thickness ranging between 80 Å to 100 Å.

(29) In one embodiment, a gate driving voltage of the NMOS upper gate device **10a** of the NMOS upper gate device region HS-NMOS is 3.3V, and the first N-type drain is electrically connected to a voltage level ranging between 6V and 36V.

(30) In one embodiment, the NMOS half-bridge power device **10** has a minimum feature size of  $0.18 \mu\text{m}$ .

(31) Note that the term “inversion current channel” means thus. Taking this embodiment as an example, when the NMOS upper gate device **10a** or the NMOS lower gate device **10b** operates in ON operation due to the voltage applied to the first gate **17a** or the second gate **17b**, an inversion layer is formed below the first gate **17a** or the second gate **17b**, so that a conduction current flows through the region of the inversion layer, which is the inverse current channel known to a person having ordinary skill in the art, so the details thereof are not redundantly explained here.

(32) Note that the term “drift current channel” means thus. Taking this embodiment as an example, the drift region provides a region where the conduction current passes through in a drifting manner when the NMOS upper gate device **10a** or the NMOS lower gate device **10b** operates in the ON

operation, and the current path through the drift region is referred to as the “drift current channel”, which is known to a person having ordinary skill in the art, so the details thereof are not redundantly explained here.

(33) Note that the top surface **11a** as referred to does not mean a completely flat plane but refers to the surface of the semiconductor layer **11'**. In the present embodiment, for example, a part of the top surface **11a** where the insulation region **12** is in contact with has a recessed portion.

(34) Note that the term “gate” in the definition of this specification refers to a semiconductor structure which includes a conductive layer, a dielectric layer, and a spacer layer. More specifically, each of the gate **17a** and the gate **17b** includes a conductive layer which is conductive, a dielectric layer in contact with the top surface **11a**, and a spacer layer which is electrically insulative. The conductive layer serves as an electrical contact of the corresponding first gate **17a** or the corresponding second gate **17b**, and is formed on and is in contact with the dielectric layer. The spacer layer is formed out of two sides of the conductive layer, as an electrical insulative layer of the corresponding first gate **17a** or the corresponding second gate **17b**.

(35) Note that the above-mentioned “N-type” and “P-type” mean that impurities of corresponding conductivity types are doped in regions of the NMOS half-bridge power device (for example but not limited to the aforementioned first N-type high voltage well **14a** and second N-type high voltage well **14b**, the aforementioned first P-type high voltage well **15a** and second N-type high voltage well **15b**, the aforementioned first N-type source **18a** and first N-type drain **18b**, and the aforementioned second N-type source **19a** and second N-type drain **19b**, etc.), so that the regions have the corresponding “N-type” or “P-type”, wherein “N-type” has conductivity type opposite to “P-type”.

(36) In addition, in the NMOS half-bridge power device, a lateral length of the drift region can be determined according to an operation voltage that the NMOS half-bridge power device is required to withstand in a normal operation, so that the NMOS half-bridge power device can operate at or higher than a specific voltage.

(37) In addition, the term “power device” refers to a device configured to deliver power, wherein when the power device is constituted by a transistor (which is usually a high voltage device), a voltage applied to the drain thereof in normal operation is higher than a specific voltage, such as 5V.

(38) Please refer to FIG. 2, which shows a cross-section view of an NMOS half-bridge power device **20** according to an embodiment of the present invention. This embodiment shown in FIG. 2 is different from the embodiment shown in FIG. 1 in that: the NMOS half-bridge power device **20** further comprises a second N-type buried layer **23a**, a third N-type high voltage isolation region **24c** and a fourth N-type high voltage isolation region **24d**.

(39) The second N-type buried layer **23a** is formed in the NMOS lower gate device region LS-NMOS by one same ion implantation process that forms the first N-type buried layer **13a**. The second N-type buried layer **23a** is formed in the semiconductor layer **11'** and in the substrate **11** and is vertically below the second N-type high voltage well **14b** and the second P-type high voltage well **15b**.

(40) The third N-type high voltage isolation region **24c** and the fourth N-type high voltage isolation region **24d** are formed in the semiconductor layer **11'** of the NMOS lower gate device region LS-NMOS by the one same ion implantation process that forms the first N-type high voltage isolation region **14c** and the second N-type high voltage isolation region **14d**.

(41) In the channel direction, the third N-type high voltage isolation region **24c** is in contact with a side of the second N-type high voltage well **14b**, wherein this side of the second N-type high voltage well **14b** is opposite to another side of the second N-type high voltage well **14b** which is in contact with the second P-type high voltage well **15b**. In the channel direction, the fourth N-type high voltage isolation region **24d** is in contact with a side of the second P-type high voltage well **15b**, wherein this side of the second P-type high voltage well **15b** is opposite to another side of the

second P-type high voltage well **15b** which is in contact with the second N-type high voltage well **14b**.

(42) The second N-type buried layer **23a**, the third N-type high voltage isolation region **24c** and the fourth N-type high voltage isolation region **24d** together constitute a device isolation region, wherein such device isolation region entirely encompasses the lower boundary of the NMOS upper gate device **10b** below the top surface **11a** of the semiconductor layer **11'**, so that below the top surface **11a** of the semiconductor layer **11'**, the NMOS upper gate device **10b** is electrically isolated from other devices which are formed in the semiconductor layer **11'**, such as the NMOS lower gate device **10a**.

(43) The first N-type buried layer **13a** and the second N-type buried layer **23a**, can be formed by, for example but not limited to, anion implantation process, which implants N conductivity type impurities in the substrate **11** in the form of accelerated ions. Next, during or subsequent to the formation of the semiconductor layer **11'**, the first N-type buried layer **13a** and the second N-type buried layer **23a** are formed by thermal diffusion.

(44) Please refer to FIG. 3A to FIG. 3N, which show cross-section views of a manufacturing method of an NMOS half-bridge power device **20** according to an embodiment of the present invention. The NMOS half-bridge power device **20** includes: an NMOS upper gate device **20a** and an NMOS lower gate device **20b**. As shown in FIG. 3A, a substrate **11** is provided. And, a first N-type buried layer **13a** and a second N-type buried layer **23a** are formed, which can be formed by, for example but not limited to, an ion implantation process, which implants N conductivity type impurities in the substrate **11** in the form of accelerated ions. Later, during or subsequent to the formation of a semiconductor layer **11'** (as shown in FIG. 3B), the first N-type buried layer **23a** and the second N-type buried layer **23b** are subject to thermal diffusion to be completely formed.

(45) Next, referring to FIG. 3B, a semiconductor layer **11'** is formed on the substrate **11**. The semiconductor layer **11'**, for example, is formed on the substrate **11** by an epitaxial process, or is a part of the substrate **11**. As described above, during or subsequent to the formation of the semiconductor layer **11'**, the first N-type buried layer **13a** and the second N-type buried layer **23a** thermally diffuse to be completely formed. The semiconductor layer **11'** has a top surface **11a** and a bottom surface **11b** opposite to the top surface **11a** in a vertical direction (as indicated by the direction of the solid arrow in FIG. 3B). The semiconductor layer **11'** can be formed by various methods known to a person having ordinary skill in the art, so the details thereof are not redundantly explained here. The substrate **11** can be for example a P-type or an N-type semiconductor substrate.

(46) Next, referring to FIG. 3C, plural isolation regions **12** is formed by one same process. The isolation regions **12** can be for example but not limited to a shallow trench isolation (STI) structure. The insulation regions **12** are formed on the semiconductor layer **11'**, for defining an NMOS upper gate device region HS-NMOS and an NMOS lower gate device region LS-NMOS, wherein the NMOS upper gate device **20a** is formed in the NMOS upper gate device region HS-NMOS, whereas, the NMOS lower gate device **20b** is formed in the NMOS lower gate device region LS-NMOS.

(47) Next, referring to FIG. 3D, the first N-type high voltage isolation region **14c**, the second N-type high voltage isolation region **14d**, the third N-type high voltage isolation region **24c** and the fourth N-type high voltage isolation region **24d** are formed in the semiconductor layer **11'** of the NMOS upper gate device region HS-NMOS by one same ion implantation process. In a channel direction (as indicated by the direction of the dashed arrow shown in FIG. 3D), the first N-type high voltage isolation region **14c** is in contact with a side of the first N-type high voltage well **14a**, wherein this side of the first N-type high voltage well **14a** is opposite to another side of the first N-type high voltage well **14a** which is in contact with the first P-type high voltage well **15a**. In the channel direction, the second N-type high voltage isolation region **14d** is in contact with a side of the first P-type high voltage well **15a**, wherein this side of the first P-type high voltage well **15a** is

opposite to another side of the first P-type high voltage well **15a** which is in contact with the first N-type high voltage well **14a**. The first N-type buried layer **13a**, the first N-type high voltage isolation region **14c** and the second N-type high voltage isolation region **14d** together constitute a device isolation region, wherein such device isolation region entirely encompasses the lower boundary of the NMOS upper gate device **20a** below the top surface **11a** of the semiconductor layer **11'**, so that below the top surface **11a** of the semiconductor layer **11'**, the NMOS upper gate device **20a** is electrically isolated from other devices which are formed in the semiconductor layer **11'**, such as the NMOS lower gate device **20b**. The second N-type buried layer **23a**, the third N-type high voltage isolation region **24c** and the fourth N-type high voltage isolation region **24d** together constitute a device isolation region, wherein such device isolation region entirely encompasses the lower boundary of the NMOS upper gate device **20b** below the top surface **11a** of the semiconductor layer **11'**, so that below the top surface **11a** of the semiconductor layer **11'**, the NMOS upper gate device **20b** is electrically isolated from other devices which are formed in the semiconductor layer **11'**, such as the NMOS lower gate device **20a**.

(48) Next, referring to FIG. 3E, the first N-type high voltage well **14a** and the second N-type high voltage well **14b** are formed in the semiconductor layer **11'** of the NMOS upper gate device region HS-NMOS and the semiconductor layer **11'** of the NMOS lower gate device region LS-NMOS, respectively, by one same ion implantation process. Both the first N-type high voltage well **14a** and the second N-type high voltage well **14b** are below and in contact with the top surface **11a**. A part of the first N-type high voltage well **14a** is located vertically below and in contact with the first gate **17a**, which serve as a drift current channel in an ON operation of the NMOS upper gate device **20a**. Besides, a part of the second N-type high voltage well **14b** is located vertically below and in contact with the second gate **17b**, which serve as a drift current channel in an ON operation of the NMOS lower gate device **20b**.

(49) Next, referring to FIG. 3F, the first P-type high voltage well **15a** and the second P-type high voltage well **15b** are formed in the semiconductor layer **11'** of the NMOS upper gate device region HS-NMOS and the semiconductor layer **11'** of the NMOS lower gate device region LS-NMOS, respectively, by one same ion implantation process. The first N-type high voltage well **14a** and the first P-type high voltage well **15a** are in contact with each other in the channel direction. The second N-type high voltage well **14b** and the second P-type high voltage well **15b** are in contact with each other in the channel direction. Both the first P-type high voltage well **15a** and the second P-type high voltage well **15b** are below and in contact with the top surface **11a**.

(50) Next, referring to FIG. 3G, a drift oxide layer **16** is formed on the semiconductor layer **11'** by for example but not limited to a deposition process, wherein the drift oxide layer **16** overlays the NMOS upper gate device region HS-NMOS and the NMOS lower gate device region LS-NMOS.

(51) Next, referring to FIG. 3H, the first drift oxide region **16a** and the second drift oxide region **16b** are formed, by one same etching process including etching a drift oxide layer, in the NMOS upper gate device region HS-NMOS and in the NMOS lower gate device region LS-NMOS, respectively. The first drift oxide region **16a** and the second drift oxide region **16b** are formed on the semiconductor layer **11'**, and are located a drift region of the NMOS upper gate device **20a** and a drift region of the NMOS lower gate device **20b**, respectively.

(52) Next, referring to FIG. 3I, subsequent to the formation of the first drift oxide region **16a** and the second drift oxide region **16b**, a gate dielectric layer **17'** is formed on the semiconductor layer **11'**, wherein the gate dielectric layer **17'** overlays the NMOS upper gate device region HS-NMOS and the NMOS lower gate device region LS-NMOS.

(53) Next, referring to FIG. 3J, subsequent to the formation of the gate dielectric layer **17'**, a polysilicon layer **17** is formed on the gate dielectric layer **17'** by for example but not limited to a deposition process, wherein the polysilicon layer **17** overlays the NMOS upper gate device region HS-NMOS and the NMOS lower gate device region LS-NMOS.

(54) Next, referring to FIG. 3K, subsequent to the formation of the polysilicon layer **17**, the

polysilicon layer **17** is etched to form a first gate **17a** in the NMOS upper gate device region HS-NMOS and to form a second gate **17b** in the NMOS lower gate device region LS-NMOS by one same etching process.

(55) Note that the thickness of the gate dielectric layer **17'** is significantly thinner than the thickness of the polysilicon layer **17**. Subsequent to the formation of the first gate **17a** and the second gate **17b**, the gate dielectric layer **17'** serves to function as a dielectric layer of the first gate **17a** and a dielectric layer of the second gate **17b**. A transistor device is known to a person having ordinary skill in the art, so the details thereof are not redundantly explained here.

(56) The first gate **17a** and the second gate **17b** are formed on the top surface **11a** of the semiconductor layer **11'**. Each of the first gate **17a** and the second gate **17b** includes: a conductive layer, a spacer layer and a dielectric layer, wherein the dielectric layer is located on and in contact with the top surface **11a**, which is known to a person having ordinary skill in the art, so the details thereof are not redundantly explained here.

(57) Next, referring to FIG. 3L, the first P-type body region **19a** and the second P-type body region **19b** are formed in the semiconductor layer **11'** of the NMOS upper gate device region HS-NMOS and the semiconductor layer **11'** of the NMOS lower gate device region LS-NMOS, respectively, by one same ion implantation process. The first P-type body region **19a** and the first N-type high voltage well **14a** are in contact with each other in the channel direction. The second P-type body region **19b** and the second N-type high voltage well **14b** are in contact with each other in the channel direction.

(58) A portion of the first P-type body region **19a** is located vertically below and in contact with the first gate **17a**, which serve as an inversion current channel in an ON operation of the NMOS upper gate device **20a**. And, a portion of the second P-type body region **19b** is located vertically below and in contact with the second gate **17b**, which serve as an inversion current channel in an ON operation of the NMOS lower gate device **10b**.

(59) Next, referring to FIG. 3M, the first N-type source **18a**, the first N-type drain **18b**, the second N-type source **18c** and the second N-type drain **18d** are formed, by one same ion implantation process, in the semiconductor layer **11'** of the NMOS upper gate device region HS-NMOS, wherein the first N-type source **18a** and the first N-type drain **18b** are located below and outside two sides of the first gate **17a** in the channel direction, respectively, wherein the side of the first gate **17a** which is closer to the first N-type source **18a** is a source side and the side of the first gate **17a** which is closer to the first N-type drain **18b** is a drain side, and wherein the first N-type source **18a** is located in the first P-type body region **19a**, and the first N-type drain **18b** is located in the first N-type high voltage well **14a**.

(60) In the vertical direction, the first N-type source **18a** and the first N-type drain **18b** are formed on and in contact with the top surface **11a**. Besides, in the channel direction, a drift region of the NMOS upper gate device **20a** lies between the first N-type drain **18b** and the first P-type body region **19a**, so as to separate the first N-type drain **18b** from the first P-type body region **19a**. And, a position in the first N-type high voltage well **14a** which is near the top surface **11a** serves as a drift current channel in an ON operation of the NMOS upper gate device **20a**.

(61) The second N-type source **18c** and the second N-type drain **18d** are located below and outside two sides of the second gate **17b** in the channel direction, respectively, wherein the side of the second gate **17b** which is closer to the second N-type source **18c** is a source side and the side of the second gate **17b** which is closer to the second N-type drain **18d** is a drain side, and wherein the second N-type source **18c** is located in the second P-type body region **19b**, and the second N-type drain **18d** is located in the second N-type high voltage well **14b**.

(62) In the vertical direction, the second N-type source **18c** and the second N-type drain **18d** are formed on and in contact with the top surface **11a**. Besides, in the channel direction, a drift region of the NMOS lower gate device **20b** lies between the second N-type drain **18d** and the second P-type body region **19b**, so as to separate the second N-type drain **18d** from the second P-type body

region **19b**. And, a position in the second N-type high voltage well **14b** which is near the top surface **11a** serves as a drift current channel in an ON operation of the NMOS lower gate device **20b**.

(63) Next, referring to FIG. **3N**, the first P-type conductive region **19c** and the second P-type conductive region **19d** are formed by one same ion implantation process. The first P-type conductive region **19c** is formed in the first P-type body region **19a**, wherein the first P-type conductive region **19c** serves as an electrical contact of the first P-type body region **19a**. The second P-type conductive region **19d** is formed in the second P-type body region **19c**, wherein the second P-type conductive region **19c** serves as an electrical contact of the second P-type body region **19b**.

(64) The first P-type conductive region **19c** and the second P-type conductive region **19d** are formed on and in contact with the top surface **11a**. Besides, in the channel direction, the first P-type conductive region **19c** is in contact with and electrically connected to the first N-type source **18a**, whereas, the second P-type conductive region **19d** is in contact with and electrically connected to the second N-type source **18c**. As shown in FIG. **1**, the first P-type conductive region **19c**, the first N-type source **18a** and the second N-type drain **18d** are electrically connected to a switch node voltage SW. In addition, the second P-type conductive region **19d** and the second N-type source **18c** are electrically connected to ground. The first N-type drain **18b** is electrically connected to an input voltage  $V_{in}$ . In a case wherein the NMOS half-bridge power device is applied to a buck power stage circuit, a switch node is electrically connected to one end of an inductor, whereas, another end of the inductor is electrically connected to an output voltage. A buck power stage circuit is known to a person having ordinary skill in the art, so the details thereof are not redundantly explained here.

(65) In one embodiment, the first gate **17a** of the NMOS upper gate device **10a** has a gate length of  $0.4 \sim 1 \mu\text{m}$ . And, a length of a portion where the first gate **17a** overlays the first drift oxide region **16a** is  $0.1 \sim 0.4 \mu\text{m}$ .

(66) In one embodiment, the second gate **17b** of the NMOS lower gate device **10b** has a gate length of  $0.4 \sim 1 \mu\text{m}$ . And, a length of a portion where the second gate **17b** overlays the second drift oxide region **16b** is  $0.1 \sim 0.4 \mu\text{m}$ .

(67) In one embodiment, the semiconductor layer **11'** is a P-type semiconductor epitaxial layer having a volume resistivity of 45 Ohm-cm.

(68) In one embodiment, each of the first drift oxide region **16a** and the second drift oxide region **16b** is a chemical vapor deposition (CVD) oxide region.

(69) In one embodiment, each of the first drift oxide region **16a** and the second drift oxide region **16b** has a thickness ranging between 400 Å to 450 Å.

(70) In one embodiment, each of the dielectric layer of the first gate **17a** and the dielectric layer of the second gate **17b** has a thickness ranging between 80 Å to 100 Å.

(71) In one embodiment, a gate driving voltage of the NMOS upper gate device **10a** of the NMOS upper gate device region HS-NMOS is 3.3V, and the first N-type drain is electrically connected to a voltage level ranging between 6V and 36V.

(72) In one embodiment, the NMOS half-bridge power device **10** has a minimum feature size of  $0.18 \mu\text{m}$ .

(73) The present invention has been described in considerable detail with reference to certain preferred embodiments thereof. It should be understood that the description is for illustrative purpose, not for limiting the broadest scope of the present invention. An embodiment or a claim of the present invention does not need to achieve all the objectives or advantages of the present invention. The title and abstract are provided for assisting searches but not for limiting the scope of the present invention. Those skilled in this art can readily conceive variations and modifications within the spirit of the present invention. For example, other process steps or structures, such as a lightly doped drain (LDD) may be added. For another example, the lithography process is not

limited to the mask technology but it can also include electron beam lithography. It is not limited for each of the embodiments described hereinbefore to be used alone; under the spirit of the present invention, two or more of the embodiments described hereinbefore can be used in combination. For example, two or more of the embodiments can be used together, or, a part of one embodiment can be used to replace a corresponding part of another embodiment. In view of the foregoing, the spirit of the present invention should cover all such and other modifications and variations, which should be interpreted to fall within the scope of the following claims and their equivalents.

## Claims

1. An NMOS half-bridge power device, comprising: a semiconductor layer, which is formed on a substrate; a plurality of insulation regions, which are formed on the semiconductor layer, for defining an NMOS upper gate device region and an NMOS lower gate device region, wherein an NMOS upper gate device is formed in the NMOS upper gate device region, whereas, an NMOS lower gate device is formed in the NMOS lower gate device region; a first N-type buried layer, which is formed in the NMOS upper gate device region; a first N-type high voltage isolation region and a second N-type high voltage isolation region, which are formed in the semiconductor layer of the NMOS upper gate device region by one same ion implantation process; a first N-type high voltage well and a second N-type high voltage well, which are formed in the semiconductor layer of the NMOS upper gate device region and the semiconductor layer of the NMOS lower gate device region, respectively, by one same ion implantation process; a first P-type high voltage well and a second P-type high voltage well, which are formed in the semiconductor layer of the NMOS upper gate device region and the semiconductor layer of the NMOS lower gate device region, respectively, by one same ion implantation process; wherein the first N-type high voltage well and the first P-type high voltage well are in contact with each other in a channel direction, and wherein the second N-type high voltage well and the second P-type high voltage well are in contact with each other in the channel direction; a first drift oxide region and a second drift oxide region, which are formed, by one same etching process including etching a drift oxide layer, in the NMOS upper gate device region and in the NMOS lower gate device region, respectively; a first gate and a second gate, which are formed, by one same etching process including etching a polysilicon layer, in the NMOS upper gate device region and in the NMOS lower gate device region, respectively; a first P-type body region and a second P-type body region, which are formed in the semiconductor layer of the NMOS upper gate device region and the semiconductor layer of the NMOS lower gate device region, respectively, by one same ion implantation process; wherein a portion of the first P-type body region is located vertically below the first gate, and wherein the first P-type body region and the first N-type high voltage well are in contact with each other in the channel direction, wherein a portion of the second P-type body region is located vertically below the second gate, and wherein the second P-type body region and the second N-type high voltage well are in contact with each other in the channel direction; a first N-type source and a first N-type drain, which are formed, by one same ion implantation process, in the semiconductor layer of the NMOS upper gate device region, wherein the first N-type source and the first N-type drain are located below and outside two sides of the first gate, respectively, wherein a side of the first gate which is closer to the first N-type source is a source side and another side of the first gate which is closer to the first N-type drain is a drain side, and wherein the first N-type source is located in the first P-type body region, and the first N-type drain is located in the first N-type high voltage well; and a second N-type source and a second N-type drain, which are formed in the semiconductor layer of the NMOS lower gate device region by the one same ion implantation process that forms the first N-type source and the first N-type drain, wherein the second N-type source and the second N-type drain are located below and outside two sides of the second gate, respectively, wherein a side of the second gate which is closer to the second N-type source is a source side and another side of the second gate which is closer to



the second N-type drain is a drain side, and wherein the second N-type source is located in the second P-type body region, and the second N-type drain is located in the second N-type high voltage well; wherein the first N-type buried layer is formed in the semiconductor layer and in the substrate and is vertically below the first N-type high voltage well and the first P-type high voltage well; wherein in the channel direction, the first N-type high voltage isolation region is in contact with a side of the first N-type high voltage well, wherein this side of the first N-type high voltage well is opposite to another side of the first N-type high voltage well which is in contact with the first P-type high voltage well; wherein in the channel direction, the second N-type high voltage isolation region is in contact with a side of the first P-type high voltage well, wherein this side of the first P-type high voltage well is opposite to another side of the first P-type high voltage well which is in contact with the first N-type high voltage well.

2. The NMOS half-bridge power device of claim 1, further comprising: a first P-type conductive region, which is formed in the first P-type body region, wherein the first P-type conductive region serves as an electrical contact of the first P-type body region; and a second P-type conductive region, which is formed in the second P-type body region by the one same ion implantation process that forms the first P-type conductive region, wherein the second P-type conductive region serves as an electrical contact of the second P-type body region.

3. The NMOS half-bridge power device of claim 2, further comprising: a second N-type buried layer, which is formed in the NMOS lower gate device region; and a third N-type high voltage isolation region and a fourth N-type high voltage isolation region, which are formed in the semiconductor layer of the NMOS lower gate device region by the one same ion implantation process that forms the first N-type high voltage isolation region and the second N-type high voltage isolation region; wherein the second N-type buried layer is formed in the semiconductor layer and in the substrate and is vertically below the second N-type high voltage well and the second P-type high voltage well; wherein in the channel direction, the third N-type high voltage isolation region is in contact with a side of the second N-type high voltage well, wherein this side of the second N-type high voltage well is opposite to another side of the second N-type high voltage well which is in contact with the second P-type high voltage well; wherein in the channel direction, the fourth N-type high voltage isolation region is in contact with a side of the second P-type high voltage well, wherein this side of the second P-type high voltage well is opposite to another side of the second P-type high voltage well which is in contact with the second N-type high voltage well.

4. The NMOS half-bridge power device of claim 2, wherein the first N-type source is electrically connected to the second N-type source, and wherein the first P-type conductive region is electrically connected to the second N-type source.

5. The NMOS half-bridge power device of claim 1, wherein the first gate of the NMOS upper gate device has a gate length of  $0.4 \sim 1 \mu\text{m}$ , and wherein a length of a portion where the first gate overlays the first drift oxide region is  $0.1 \sim 0.4 \mu\text{m}$ .

6. The NMOS half-bridge power device of claim 1, wherein the second gate of the NMOS lower gate device has a gate length of  $0.4 \sim 1 \mu\text{m}$ , and wherein a length of a portion where the second gate overlays the second drift oxide region is  $0.1 \sim 0.4 \mu\text{m}$ .

7. The NMOS half-bridge power device of claim 1, wherein the semiconductor layer is a P-type semiconductor epitaxial layer having a volume resistivity of  $45 \text{ Ohm}\cdot\text{cm}$ .

8. The NMOS half-bridge power device of claim 1, wherein each of the first drift oxide region and the second drift oxide region has a thickness ranging between  $400 \text{ \AA}$  and  $450 \text{ \AA}$ .

9. The NMOS half-bridge power device of claim 1, wherein each of the dielectric layer of the first gate and the dielectric layer of the second gate has a thickness ranging between  $80 \text{ \AA}$  and  $100 \text{ \AA}$ .

10. The NMOS half-bridge power device of claim 1, wherein a gate driving voltage of the NMOS upper gate device region is  $3.3\text{V}$ , and wherein the first N-type drain is electrically connected to a voltage level ranging between  $6\text{V}$  and  $36\text{V}$ .

11. The NMOS half-bridge power device of claim 1, wherein the NMOS half-bridge power device

has a minimum feature size of 0.18  $\mu\text{m}$ .

12. A manufacturing method of an NMOS half-bridge power device, wherein the NMOS half-bridge power device includes: an NMOS upper gate device and an NMOS lower gate device; the manufacturing method of the NMOS half-bridge power device comprising steps of: forming a semiconductor layer on a substrate; forming a plurality of insulation regions on the semiconductor layer, for defining an NMOS upper gate device region and an NMOS lower gate device region, wherein an NMOS upper gate device is formed in the NMOS upper gate device region, whereas, an NMOS lower gate device is formed in the NMOS lower gate device region; forming a first N-type buried layer in the NMOS upper gate device region; forming a first N-type high voltage isolation region and a second N-type high voltage isolation region in the semiconductor layer of the NMOS upper gate device region by one same ion implantation process; forming a first N-type high voltage well and a second N-type high voltage well in the semiconductor layer of the NMOS upper gate device region and the semiconductor layer of the NMOS lower gate device region, respectively, by one same ion implantation process; forming a first P-type high voltage well and a second P-type high voltage well in the semiconductor layer of the NMOS upper gate device region and the semiconductor layer of the NMOS lower gate device region, respectively, by one same ion implantation process; wherein the first N-type high voltage well and the first P-type high voltage well are in contact with each other in a channel direction, and wherein the second N-type high voltage well and the second P-type high voltage well are in contact with each other in the channel direction; forming a drift oxide layer on the semiconductor layer, wherein the drift oxide layer overlays the NMOS upper gate device region and the NMOS lower gate device region; etching the drift oxide layer to form a first drift oxide region in the NMOS upper gate device region and to form a second drift oxide region in the NMOS lower gate device region by one same etching process; subsequent to the formation of the first drift oxide region and the second drift oxide region, forming a gate dielectric layer on the semiconductor layer, wherein the gate dielectric layer overlays the NMOS upper gate device region and the NMOS lower gate device region; forming a polysilicon layer on the gate dielectric layer, wherein the polysilicon layer overlays the NMOS upper gate device region and the NMOS lower gate device region; etching the polysilicon layer to form a first gate in the NMOS upper gate device region and to form a second gate in the NMOS lower gate device region by one same etching process; forming a first P-type body region and a second P-type body region in the semiconductor layer of the NMOS upper gate device region and the semiconductor layer of the NMOS lower gate device region, respectively, by one same ion implantation process; wherein a portion of the first P-type body region is located vertically below the first gate, and wherein the first P-type body region and the first N-type high voltage well are in contact with each other in the channel direction, wherein a portion of the second P-type body region is located vertically below the second gate, and wherein the second P-type body region and the second N-type high voltage well are in contact with each other in the channel direction; forming a first N-type source and a first N-type drain in the semiconductor layer of the NMOS upper gate device region by one same ion implantation process, wherein the first N-type source and the first N-type drain are located below and outside two sides of the first gate, respectively, wherein a side of the first gate which is closer to the first N-type source is a source side and another side of the first gate which is closer to the first N-type drain is a drain side, and wherein the first N-type source is located in the first P-type body region, and the first N-type drain is located in the first N-type high voltage well; and forming a second N-type source and a second N-type drain in the semiconductor layer of the NMOS lower gate device region by the one same ion implantation process that forms the first N-type source and the first N-type drain, wherein the second N-type source and the second N-type drain are located below and outside two sides of the second gate, respectively, wherein a side of the second gate which is closer to the second N-type source is a source side and another side of the second gate which is closer to the second N-type drain is a drain side, and wherein the second N-type source is located in the second P-type body region, and the

second N-type drain is located in the second N-type high voltage well; wherein the first N-type buried layer is formed in the semiconductor layer and in the substrate and is vertically below the first N-type high voltage well and the first P-type high voltage well; wherein in the channel direction, the first N-type high voltage isolation region is in contact with a side of the first N-type high voltage well, wherein this side of the first N-type high voltage well is opposite to another side of the first N-type high voltage well which is in contact with the first P-type high voltage well; wherein in the channel direction, the second N-type high voltage isolation region is in contact with a side of the first P-type high voltage well, wherein this side of the first P-type high voltage well is opposite to another side of the first P-type high voltage well which is in contact with the first N-type high voltage well.

13. The manufacturing method of the NMOS half-bridge power device of claim 12, further comprising: forming a first P-type conductive region in the first P-type body region, wherein the first P-type conductive region serves as an electrical contact of the first P-type body region; and forming a second P-type conductive region in the second P-type body region by the one same ion implantation process that forms the first P-type conductive region, wherein the second P-type conductive region serves as an electrical contact of the second P-type body region.

14. The manufacturing method of the NMOS half-bridge power device of claim 13, further comprising: forming a second N-type buried layer in the NMOS lower gate device region; and forming a third N-type high voltage isolation region and a fourth N-type high voltage isolation region in the semiconductor layer of the NMOS lower gate device region by the one same ion implantation process that forms the first N-type high voltage isolation region and the second N-type high voltage isolation region; wherein the second N-type buried layer is formed in the semiconductor layer and in the substrate and is vertically below the second N-type high voltage well and the second P-type high voltage well; wherein in the channel direction, the third N-type high voltage isolation region is in contact with a side of the second N-type high voltage well, wherein this side of the second N-type high voltage well is opposite to another side of the second N-type high voltage well which is in contact with the second P-type high voltage well; wherein in the channel direction, the fourth N-type high voltage isolation region is in contact with a side of the second P-type high voltage well, wherein this side of the second P-type high voltage well is opposite to another side of the second P-type high voltage well which is in contact with the second N-type high voltage well.

15. The manufacturing method of the NMOS half-bridge power device of claim 13, wherein the first N-type source is electrically connected to the second N-type source, and wherein the first P-type conductive region is electrically connected to the second N-type source.

16. The manufacturing method of the NMOS half-bridge power device of claim 12, wherein the first gate of the NMOS upper gate device has a gate length of  $0.4 \sim 1 \mu\text{m}$ , and wherein a length of a portion where the first gate overlays the first drift oxide region is  $0.1 \sim 0.4 \mu\text{m}$ .

17. The manufacturing method of the NMOS half-bridge power device of claim 12, wherein the second gate of the NMOS lower gate device has a gate length of  $0.4 \sim 1 \mu\text{m}$ , and wherein a length of a portion where the second gate overlays the second drift oxide region is  $0.1 \sim 0.4 \mu\text{m}$ .

18. The manufacturing method of the NMOS half-bridge power device of claim 12, wherein the semiconductor layer is a P-type semiconductor epitaxial layer having a volume resistivity of  $45 \text{ Ohm-cm}$ .

19. The manufacturing method of the NMOS half-bridge power device of claim 12, wherein each of the first drift oxide region and the second drift oxide region has a thickness ranging between  $400 \text{ \AA}$  and  $450 \text{ \AA}$ .

20. The manufacturing method of the NMOS half-bridge power device of claim 12, wherein each of the dielectric layer of the first gate and the dielectric layer of the second gate has a thickness ranging between  $80 \text{ \AA}$  and  $100 \text{ \AA}$ .

21. The manufacturing method of the NMOS half-bridge power device of claim 12, wherein a gate

driving voltage of the NMOS upper gate device region is 3.3V, and wherein the first N-type drain is electrically connected to a voltage level ranging between 6V and 36V.

22. The manufacturing method of the NMOS half-bridge power device of claim 12, wherein the NMOS half-bridge power device has a minimum feature size of 0.18  $\mu\text{m}$ .

---